MILITARY SPECIFICATION

SEALING COMPOUND, ALUMINUM STRUCTURE, PRESSURE AND WEATHER SEALING, LOW DENSITY

1. SCOPE

1.1 Scope. This specification covers low density, activated synthetic rubber compounds for sealing and repairing in pressure and weather cavity areas in a temperature range of -67°F to +200°F. These materials are not intended for faying surface sealing or for wet installing fasteners since the fillers used to obtain low density may reduce structural fatigue strength when used to seal between metal surfaces.

1.2 Classification. The sealing compound shall be of the following classes:

Class A - Brushable.

Class B - Application by extrusion gun and spatula for filleting.

1.2.1 Dash numbers. The following dash numbers shall be used to designate the minimum application time in hours.

Class A dash numbers shall be 1/2, and 2.

Class B dash numbers shall be 1/2, 2 and 4.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-A-250/12	Aluminum Alloy 7075, Plate and Sheet
QQ-A-250/13	Aluminum Alloy Alclad 7075, Plate and Sheet

FSC 8030

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TT-N-95	Naphtha, Aliphatic
TT-S-735	Standard Test Fluids, Hydrocarbon
CCC-C-419	Cloth, Duck, Cotton, Unbleached, Plied-Yarns Army and Numbered
PPP-C-96	Can, Metal 28 Gage and Lighter

Military

MIL-S-4383	Sealing Compound, Topcoat, Fuel Tank, Buna-N Type
MIL-P-5425	Plastic Sheet, Acrylic, Heat Resistant
MIL-C-5541	Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys
MIL-P-8585	Primer Coating, Zinc Chromate, Low-Moisture Sensitivity
MIL-P-23377	Primer Coating, Epoxy Polyamide, Chemical and Solvent Resistant
MIL-P-38714	Packaging and Packing of Two Component Materials in Semkits
MIL-C-38736	Cleaning Compound, Solvent, for Use in Integral Fuel Tanks

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

PUBLICATIONS

U.S. Air Force

Specification Bulletin No. 535 Free Diisocyanate Test Method

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply:

AMERICAN SOCIETY FOR TESTING AND METALS (ASTM) STANDARDS

D412 Tension Testing of Vulcanized Rubber
D2240 Indentation of Rubber and Plastics By Means of a Durometer

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Qualification. The sealing compound furnished under this specification shall be a product which has been tested, and has passed the qualification tests specified herein, and has been listed or approved for listing on the applicable Qualified Products List.

3.2 Material. The base ingredient used in the manufacture of this product shall be synthetic rubber of the polysulfide or non-polyester polyurethane type. The sealant shall cure at room temperature upon the addition of a separate curing agent to the base compound. One part products that depend upon solvent evaporation or atmospheric moisture for curing are not satisfactory materials. A primer recommended by the manufacturer may be necessary to insure adhesion.

3.3 Appearance. The base compound and activator shall be of uniform blend and free of skins, lumps, gel or coarse particles. There shall be no separation of material that cannot be readily re-dispersed.

3.3.1 Color. The accelerator used shall possess sufficient color contrast to the base to permit easy identification of an unmixed or incompletely mixed compound. Neither the base compound nor the cured sealing compound shall be black, red or pink in color.

3.4 Toxicity. The compounds shall contain no benzene or chlorinated solvents. The vapor shall cause no discomfort or injury to workmen engaged in the application of the activated compound while working in a well ventilated area.

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3.5 Physical properties

3.5.1 Viscosity of base compound. When tested in accordance with 4.7, the viscosity of the class B base compound shall be 6,000 to 20,000 poises. The viscosity of the class A base compound shall be 200 to 700 poises.

3.5.2 Specific gravity. When tested per 4.8, the specific gravity of the cured compound shall not exceed 0.80.

3.5.3 Nonvolatile content. When tested in accordance with 4.9, the activated sealant shall have a minimum nonvolatile content as follows:

Class B	97 percent
Class A	To Be Established

3.5.4 Flow. The flow of activated class B sealants shall be 0.05 to 0.75 inches when tested in accordance with 4.10.

3.5.5 Application time. The activated class B sealant, when tested in accordance with 4.11, shall have a minimum extrusion rate of 10 grams per minute at the end of the rated application times. The material shall remain suitable for application for the respective time represented by the dash number per 1.2.1. class A sealant; when tested in accordance with 4.11, shall have a viscosity of less than 2,500 poises at the end of the rated application time.

3.5.6 Tack-free time. The activated sealant, when tested in accordance with 4.12, shall cure to a tack-free condition in not more than the time in hours specified in table I. At tack-free time, the polyethylene film shall peel cleanly from the sealant without deforming it.

3.5.7 Curing rate. The activated sealant, when tested in accordance with 4.13, shall have Type A durometer hardness of 30 minimum after curing for not more than the time specified in table I.

Table I. Tack-free time and curing rate

Class B	Tack-free Time (hours)	Curing Rate (hours)
B-1/2	To be established	To be established
B-2	24	72
B-4	To be established	To be established
A-2	To be established	To be established

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3.5.8 Peel strength. When tested in accordance with 4.14 the peel strength shall average at least 15 pounds per inch of width, for each specimen. Low values due to air bubbles, cloth failure, cutting, or load being picked up after cutting, shall be disregarded. The separation shall be a minimum of 95 percent in cohesion, disregarding areas where sealant to cloth failures occurred.

3.5.9 Low-temperature flexibility. The cured sealant, when tested in accordance with 4.15, shall show no evidence of cracking, checking, or loss of adhesion. Failure of any specimen shall be cause for rejection.

3.5.10 Tensile strength, elongation and hardness. When tested in accordance with 4.16, the average tensile strength, elongation, and hardness of the cured sealing compound shall not be less than the values shown in table II.

Table II. Tensile strength, elongation and hardness

Condition	Tensile Strength (min.)	Elongation (min.)	Rex Hardness (min.)
Control	100	250	40
30 days @ 160°F	100	200	35
30 days @ 200° ±2°F	75	100	35

3.5.11 Hydrolytic stability. After aging in accordance with 4.17, the loss in hardness shall be a maximum of 20 percent.

3.5.12 Moisture absorption. When tested in accordance with 4.18, the sealing compound shall not increase in weight more than 2.0 percent.

3.5.13 Reparability. Class B sealing materials shall be suitable for repairing minor breaks in themselves or with other qualified material. When tested in accordance with 4.19, the material shall have a peel strength not lower than 15 pounds per inch of width and shall not show unsatisfactory bonding, blistering, loss of adhesion, or other irregularities.

3.5.14 Crazing effects. Class B sealants shall not deteriorate or craze test strips of polymethyl methacrylate when tested in accordance with 4.20.

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3.5.15 Corrosion. Cured specimens of each class of sealant, when tested as specified in 4.21, shall show no visual evidence of tackiness, blistering, or other visual evidence of decomposition. In addition, the metal substrate underneath the sealant shall show no sign of being corroded.

3.5.16 Resistance to paint strippers. The cured sealing compounds, when tested per 4.22, shall show no evidence of tackiness or blistering and shall not lose more than 20 points in hardness.

3.5.17 Accelerated storage stability. When tested in accordance with 4.23, the mixed material shall meet Flow (3.5.4) and Curing Rate (3.5.7) and Application Time requirements (3.5.5). The peel strength shall be 15 pounds minimum. The aged compound shall show no skinning, hardening, separation, or settling of the material. After aging, the activator shall not be adversely affected and shall be capable of being restored to a condition suitable for use by normal agitation.

3.5.18 Free diisocyanate. The maximum amount of free diisocyanate in the sealing compound shall be 1.0 percent when tested as specified in 4.24.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the sealing compound shall be classified as follows:

a. Qualified tests (4.3).
b. Quality conformance inspection (4.4).

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4.3 Qualification tests. The first material that a manufacturer must qualify is class B-2. That material must meet all the requirements of this specification with the exception of those requirements that are for other classes. Once qualification for class B-2 is obtained, other classes may be qualified. The formulation for class A materials and other class B materials shall be the same as that for the class B-2, except for minor variations necessary for viscosity and application life. After the compound has been accepted for qualification, approval will be granted and the material will be identified by reference to the manufacturer's code or formula number.

4.3.1 Qualification test samples. The qualification test samples shall consist of 5 quarts of the compound upon which qualification is desired. The qualification samples shall be identified as specified herein and forwarded to the activity responsible for qualification as designated in the letter of authorization from that activity (see 6.4).

Samples for qualification tests:

SEALING COMPOUND, AIRCRAFT STRUCTURE, PRESSURE AND WEATHER SEALING, LOW DENSITY

Class and dash number
Specification MIL-S-83315(USAF)
Manufacturer's code number
Name of manufacture
Submitted by (name) (date) for qualification tests in accordance with MIL-S-83315(USAF) authorization (reference authorizing letter)
Mixing instructions.

4.4 Quality conformance inspection

4.4.1 Lot. A lot shall consist of all material of the same identity, from the same batch, and submitted at the same time for inspection.

4.4.2 Batch. A batch shall be the quantity of material run through a mill or mixer at one time.

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4.4.3 Sampling for quality conformance tests. An appropriate 1-quart sample shall be selected from each lot. The material shall be packaged and mixed as much as practical in the same containers that are being procured; that is, if the procurement is for packaging in accordance with MIL-P-38714, the conformance tests shall be conducted with material mixed in those packages.

4.4.4 Quality conformance tests. Quality conformance tests for acceptance of individual lots shall consist of the following tests:

a. Viscosity of base compound
b. Specific gravity
c. Nonvolatile content
d. Flow (class B only)
e. Application time
f. Tack-free time
g. Curing rate
h. Peel strength (Immerse for 5 days)

4.4.5 Test data. Suppliers shall maintain on file actual test data showing complete conformance to this specification on a minimum of two production lots of the base compound and curing agent. In addition, the supplier shall maintain on file acceptance test data for every lot manufactured and sold with a certification that the lot shipped meets the requirements of this specification.

4.5 Test conditions. Standard laboratory testing conditions shall be 77° ± 2°F and 50 ± 5 percent relative humidity. Except as otherwise specified hereinafter, all test specimens shall be prepared, cured, and tested under these conditions.

4.6 Preparation of test specimens

4.6.1 Cleaning of test panels. All test panels shall be cleaned by scrubbing and rinsing with clean gauze sponges using solvent formulated in accordance with MIL-C-38736. The panels shall be wiped dry with gauze sponges rather than allowing the solvent to evaporate. Gauze sponges shall be used for one solvent scrubbing or drying only. The solvent cleaned panels shall be primed within 8 hours of cleaning. The primer shall be applied according to the manufacturer's recommendation and in the film thickness recommended by the manufacturer.

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4.6.2 Preparation of sealing compound. The quantity of sealing compound required for the test shall be mixed as thoroughly as practicable. A Semco Model SP-1350 mixer, or equal, shall be used for class B-2 and B-4. The quantity of base compound to be mixed shall not be less than 600 or more than 700 grams. Class A-1/2 and A-2 shall be hand mixed. Class B-1/2 shall be mixed in MIL-P-38714 packages. The mixed sealing compound shall have a minimum inclusion of air. Where applicable, the sealing compound, immediately after mixing, shall be placed into cartridges for extrusion from the Semco No. 250 gun, or equal.

4.6.3 Quick freezing, Class B-2 and B-4 Material. After the B-2 and B-4 materials are loaded into the cartridges, both ends of the cartridge shall be closed off immediately after filling. The installed plunger constitutes a satisfactory plug at one end. The sealant shall be quick frozen immediately in a trichloroethylene and dry ice bath at -80°F or lower for 30 minutes. The cartridge shall be immersed with its plugged nozzle end down and the upper end about 1 inch above the liquid level. For tests on Tack-Free, Curing Rate, Application Time, and Flow, the class B-2 and B-4 material shall be stored at or below -40°F for at least 16 hours but not more than 48 hours. If frozen sealant is stored at -40°F, it shall be allowed to stabilize at -67° ± 2°F for 2 hours prior to testing. Thawout shall be accomplished by immersion of the frozen cartridge in a 120° ± 2°F water bath for 18 minutes with the plugs installed and the upper end of the cartridge 1 inch above the liquid level. Time zero shall be considered as occurring at the end of the 18 minute period and the timed tests begun. For all other tests, the storage time of the frozen material shall not exceed 10 days at -67° ± 2°F. Thawout shall be accomplished as above except sealant may be thawed to room temperature if desired.

4.6.4 Application of sealing compound. Unless otherwise specified herein, primed test panels shall be given an application of sealing compound to produce a coating having a thickness of 1/8 ± 1/64 inch when cured. Class A materials shall be built up by three applications, allowing 2 hours between applications to permit release of solvents. Unless otherwise specified, test specimens shall receive a 14-day cure at 77° ± 2°F and 50 ± 5 percent relative humidity. These conditions shall constitute a standard cure and all references to a standard cure shall mean these conditions. Tests on the cured sealing compound shall commence not later than 2 days after completion of the specified cure.

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4.7 Viscosity of base compound. Viscosity shall be determined after 24 hours storage at 77° ± 2°F, then thoroughly mixing the base compound and a 1-hour stabilizing period. The Brookfield viscometer shall be used and the reading obtained converted to poises. The following settings shall be observed:

	Model	Spindle	Speed
Class B	RVF, HBF or HBT	#7	2 rpm
Class A	RVF	#7	10 rpm

The instrument shall be run 1 minute in the material prior to taking the reading. If additional readings are taken, all conditions except the 24-hour storage period shall be repeated.

4.8 Specific gravity. Three specimens approximately 0.125 by 1 inch shall be cut from the sealant casting prepared for the Tensile Strength and Elongation Test, 4.16. After being weighed accurately to three decimal places in grams, a sinker shall be attached and the samples shall be dipped in methyl alcohol and then, while wet, be immediately suspended in distilled water at 77° ± 2°F and reweighed. The specific gravity shall be computed from the following formula:

$$\text{Specific gravity} = \frac{(a)}{(b) - (c)} \quad \text{where:} \qquad (1)$$

(a) = weight of test specimen in air
(b) = weight of sinker in water plus weight of specimen in air
(c) = weight of specimen and sinker in water

4.9 Nonvolatile content. Five to 10 grams of freshly mixed sealing compound shall be transferred as rapidly as possible to a cup approximately 3 inches in diameter and 3/4-inch in depth. A fitted cover shall immediately be placed over the cup and the weight determined accurately, using an enclosed beam balance that will weigh accurately to the nearest milligram. The weight of the cup and cover shall be determined accurately prior to using, and subtracted from the initial and final weights in order to calculate the new sample weights (see 3.3.2). The cover shall then be removed, and the sealing compound heated in an oven for 7 days at 158°F. It shall then be cooled in a desiccator, the cover replaced, and the weight determined accurately. The percentage of total nonvolatile content shall be calculated as follows:

$$\text{Percent of nonvolatile content} = \frac{\text{Final weight}}{\text{Initial weight}} \times 100 \qquad (2)$$

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4.10 Flow. The class B material shall be prepared as detailed in 4.6.2. The test shall be conducted with a Flow-Test jig as shown in figure 1. Depth of plunger tolerance is critical and shall be controlled within the tolerance during all tests.

4.10.1 The Flow-Test jig shall be placed on a table with the front face upward and with the plunger depressed to the limit of its travel. An excess of the mixed sealant material shall be extruded from the application gun with Semco #440 nozzle into the recessed cavity of the jig while minimizing entrapped air. The sealant shall be leveled off even with the block by scraping with a spatula in two passes, each starting at the center and moving toward the sides of the testing jig. Within 10 seconds after the leveling operation, the jig shall be placed on its end and the plunger immediately advanced to the limit of its forward travel. The Flow measurement shall be taken exactly 30 minutes after the sealing material has been applied to the test jig. The flow shall be measured from the tangent to the lower edge of the plunger to the farthest point to which flow has advanced.

4.10.2 As the sealing material progresses in its application time, the test jig shall be filled as described in 4.10.1 and the Flow Test shall be conducted at each time interval specified in table III.

Table III. Flow-test intervals

Minimum Specified Application Time	Time at Which Flow Jig is Filled
1/2 Hour	Time Zero*
2 Hours	Time Zero* 60 Minutes 120 Minutes
4 Hours	Time Zero* 120 Hours 180 Hours 240 Hours

* Time Zero is defined in 4.6.3.

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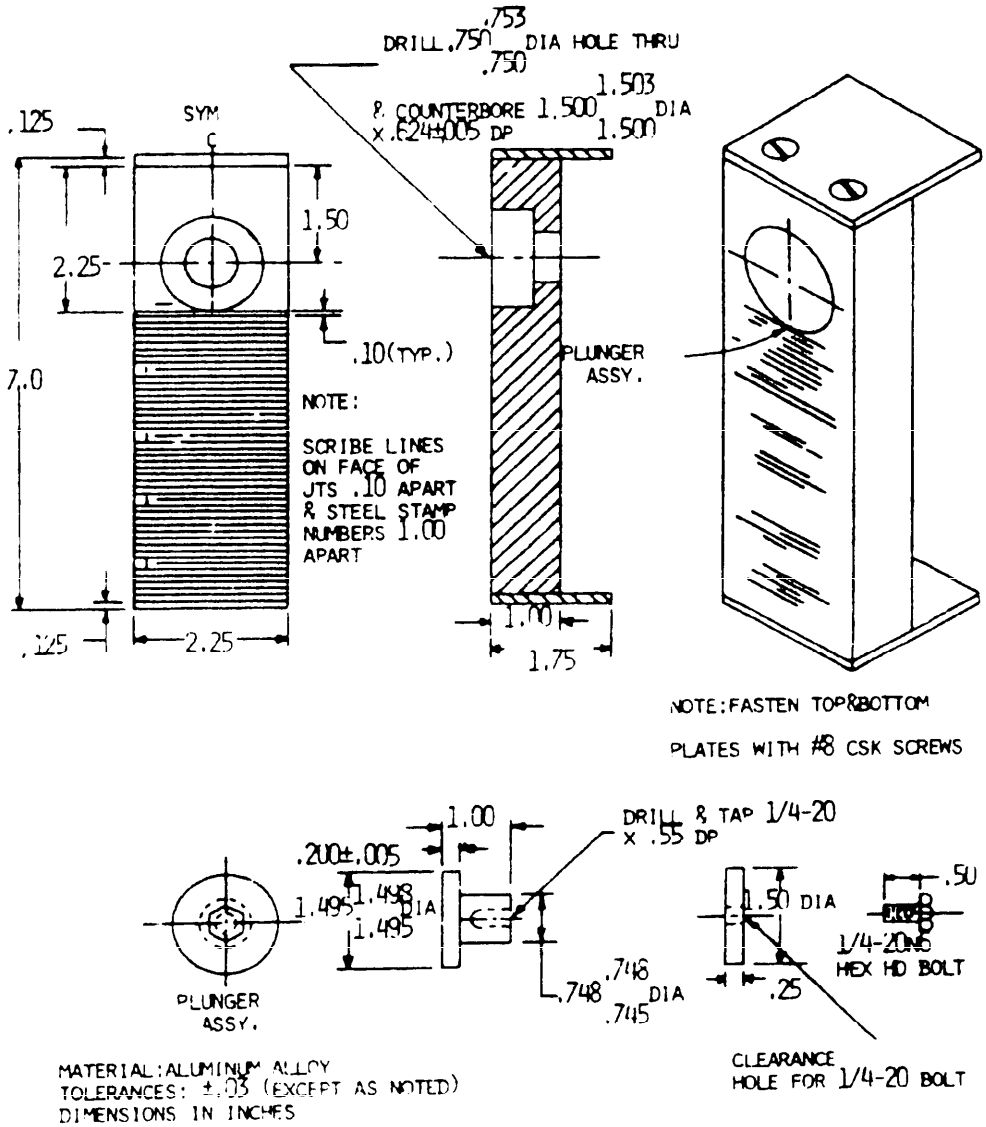


Figure 1. Flow test jig

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4.11 Application time. Class B sealant shall be prepared as described in 4.6.2. A Semco #440 nozzle with orifice diameter of 0.125 ± 0.005 inches shall be fitted onto the cartridge. The nozzle bore shall be smooth, and free of flash or other protrusions. The gun shall be attached to a constant air supply of 90 ± 2 psi. Two to three inches of sealant shall be extruded initially to clear trapped air and at least 1 inch of sealant must be extruded prior to the observation. Extrusion rate observations shall be conducted as follows: At the end of the specified application time, the sealing material shall be extruded onto a tared sheet of plastic, using the specified gun pressure, for an exactly measured time interval of 60 seconds. The extruded sealing material shall be weighed, and the grams delivered per minute recorded to the nearest gram. Class A sealant after stabilization at standard conditions for at least 6 hours, the class A sealing compound shall be mixed in the proper ratio to provide enough material to fill a pint can to within 1 inch of the top. The top should remain on the can to prevent loss of volatiles when no test is being conducted. The initial viscosity shall be determined 15 minutes after mixing using a Brookfield RVF, No. 7 spindle and 10 rpm. The final viscosity determination shall be made at the conclusion of the rated application time. One reading shall be taken after the instrument has run in the material for 1 minute.

4.12 Tack-free time. A 2.75 by 6 by 0.040 inch QQ-A-250/13 panel shall be prepared and covered with sealant to a depth of 0.125 ± 0.015 inch. The sealant shall be mixed and handled as described in 4.6.2. A piece of polyethylene film approximately 1-inch wide shall be laid across the sealing material at each end of the panel. The sealant shall be kept at 77° ± 2°F and 50 ± 5 percent relative humidity. At the end of the specified tack free time, a 0.5 by 6 inch polyethylene film (0.004 ± 0.002 inch thick) shall be applied to the sealing material using light finger pressure. Two minutes after applying the polyethylene film, the polyethylene strip shall then be slowly and evenly withdrawn at right angles from the sealant surface.

4.13 Curing rate. The curing rate shall be determined in accordance with ASTM D2240. The sealant shall be mixed and handled as described in 4.6.2. The durometer reading shall be taken on a double back-to-back 0.125 inch thick specimen for class B sealant at the time shown in table I.

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4.14 Peel strength

Clean four panels of each of the following surfaces per 4.6.1, then coat a continuous layer of sealant (0.125 ± 0.015 inch thick) on 5 inches of panel using the Peel Panel Jig shown in figure 2.

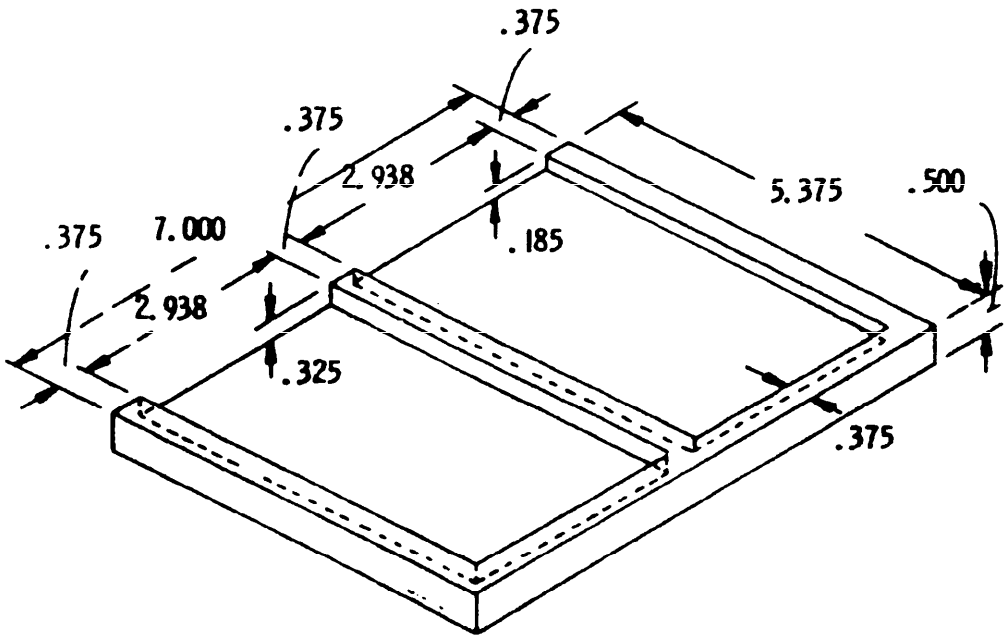
a. 0.040 by 2.75 by 6 inch aluminum alloy conforming to QQ-A-250/12 treated per MIL-C-5541 and sprayed with manufacturer's recommended primer. After the primer is cured (at room temperature), sealant shall be applied following cleaning of primer with MIL-C-38736 solvent cleaner. Primer shall cure for not more than 24 hours at standard conditions.

b. 0.040 by 2.75 by 6 inches aluminum alloy conforming to QQ-A-250/12 treated per MIL-C-5541 and sprayed with the manufacturer's recommended MIL-P-8585 primer manufacturer's instructions. Allow a 24-hour cure before cleaning panels. Clean panels with MIL-C-38736 cleaner.

c. 0.040 by 2.75 by 6 inch aluminum alloy conforming to QQ-A-250/12 treated per MIL-C-5541 and sprayed with MIL-P-23377 primer per manufacturer's instructions. Allow a 7-day cure at standard conditions before cleaning panels. Clean panels with MIL-C-38736 cleaner.

A 2-3/4 by 12-inch strip of wire screen (20- to 40-mesh aluminum or monel wire fabric) or cotton duck conforming to type III of CCC-C-419 shall be primed with the manufacturer's recommended primer, and impregnated with sealant, so that approximately 5 inches at one end is completely covered on both sides. The sealant must be worked well into the fabric. The sealant-impregnated end of the fabric shall be placed on the sealant coated panel, and smoothed down on the layer of the sealant, taking care not to trap air beneath the fabric. An additional 1/32-inch thick coating of sealing compound shall be applied over the fabric. After standard cure, two panels of each type of surface shall be completely immersed for 20 days at 140° ± 2°F in a 1-quart jar containing a 3.0 percent aqueous sodium chloride solution. (Immersion in wide-mouth quart jars with two panels in each jar is suitable.) After specified exposure at 140° ± 2°F, the panels shall be cooled in the fluid for 24 hours at standard conditions. The peel strength shall be measured within 10 minutes after removal from the test fluid. A 1-inch wide section shall be cut through the fabric and sealing compound on each panel to provide the center inch for the peel test. The specimens shall be stripped back at an angle of 180 degrees to the metal panel in a suitable tensile testing machine having a jaw separation rate of 2 inches per minute. During

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MATERIALS: ALUMINUM
DIMENSIONS IN INCHES
TOLERANCE: ± .005 UNLESS
OTHERWISE SPECIFIED.

Figure 2. Peel panel jig

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the peel strength testing, three cuts shall be made through the sealing compound to the panel in an attempt to promote adhesive failure. The cuts shall be at approximate 1-inch intervals. The results shall be the numerical average of the peak loads. Failures of the sealing compound to the fabric shall not be included in the peel strength values.

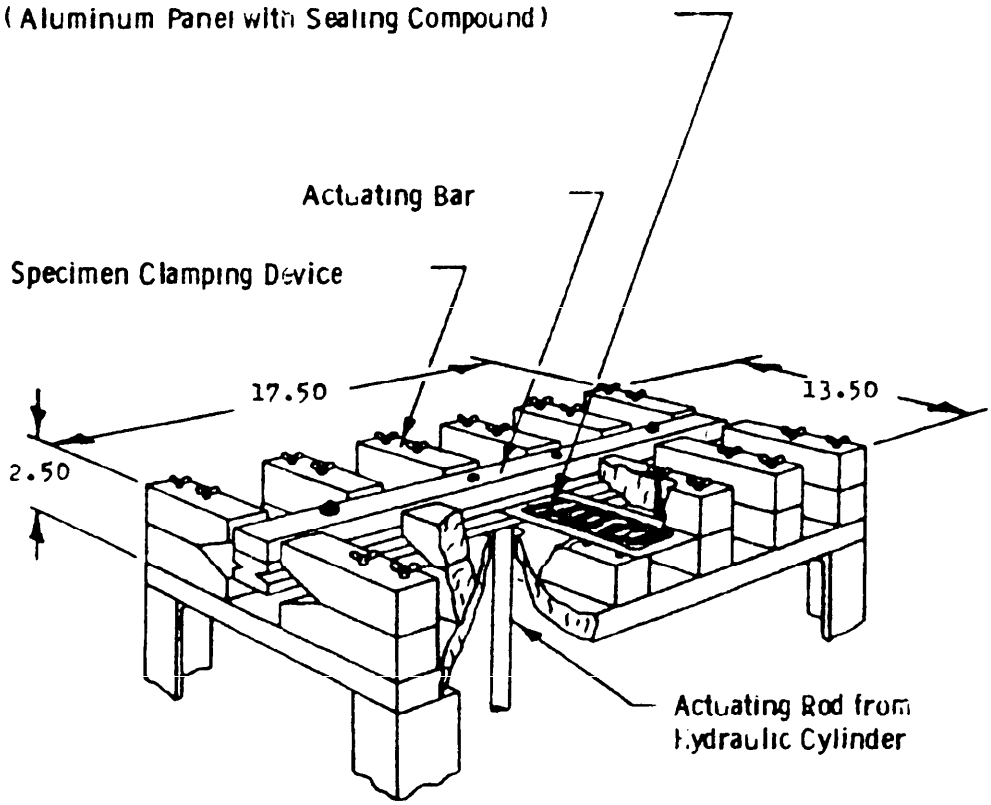
4.15 Low-temperature flexibility. Nine panels, 2.75 by 6 by .040 inches, aluminum alloy conforming to QQ-A-250/12 treated per MIL-C-5541. Three panels shall be primed with manufacturer's recommended primer, three with manufacturer's recommended MIL-P-8585 primer, and three with MIL-P-23377 primer. A coating of the sealing compound 0.094 inch thick by 1-1/2 inches wide by 4 inches long shall be applied to the center of each of the nine panels. Care shall be taken to maintain an accurate sample thickness of 0.094 inch. At the end of a 14-day cure, as specified in 4.6.4, 2 of the panels with each primer (six panels) shall be conditioned at 200° ± 2°F for an additional 7 days. All nine panels shall then be immediately placed in a low temperature flexibility jig consisting of a clamp support that will grip both sides of both 6-inch edges of the panel for a distance of 3 inches from one end without touching the sealant (see figure 3). The jig shall be capable of flexing the panel through a 30° arc (15° each side of the center) at a constant speed of 1 cycle per 5 seconds. The temperature shall be reduced to -67° ± 2°F, stabilized at this temperature for 2 hours, and the panels flexed through 130 consecutive cycles (see 3.5.9).

4.16 Tensile strength, elongation and hardness. The mixed sealant shall be cast to a thickness of 0.125 ± 0.008 inches in a closed rigid mold lined with a suitable release agent by extrusion from a sealing gun with Semco #440 nozzle. The nozzle shall be freed of air by a preliminary extrusion of 2 to 3 inches of sealant. During the casting operation, the tip of the nozzle shall be placed in contact with the lower surface of the mold and shall not be removed until the mold is filled to excess. Within 12 hours after the end of the Curing Rate time designated in table 1, the specimen shall be removed from the modl and allowed to continue curing until the standard cure time is reached (14 days). Six 6.0 by 6.0 inch specimens shall be cut from the molded slabs.

The specimens shall be exposed as follows:

a. Control
b. 30 days at 160° ± 2°F
c. 30 days at 200° ± 2°F

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DIMENSIONS IN INCHES, UNLESS OTHERWISE SPECIFIED
TOLERANCES: ± .015

Figure 3. Low-temperature flexibility jig

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After the specified exposure times, the specimens exposed to dry heat (b. and c.) shall be cooled to room temperature from 8 to 24 hours. At least four dumbbells shall be made from each specimen using Die C, ASTM D412 and then pulled on a tensile machine at the rate of 20 inches per minute. The tensile and elongation shall be recorded and averaged. Hardness shall be determined in accordance with ASTM D2240.

4.17 Hydrolytic stability. Four 1-inch squares of sealant shall be cut from the sheet cast for tensile specimens (4.16). The hardness shall be determined and each specimen shall be placed in an open aluminum dish similar to those used for nonvolatile content (4.9) and exposed to 160°F and 95 percent Relative Humidity for 120 days. At the end of this time the samples shall be removed from the aging chamber, conditioned a minimum of 4 days (and not more than 7 days) at 77°F, 50 percent Relative Humidity. Hardness shall be determined in accordance with ASTM D2240.

4.18 Moisture absorption. Three 1 by 2 inch specimens shall be cut from the casting made for tensile and elongation tests. After being weighed, the specimens shall be immersed for 20 days in distilled water at 120°F ± 2°F. Following immersion, remove the specimens, quickly wipe dry with a soft cloth and weigh. Determine weight increase as follows:

$$\frac{\text{Final Weight - Initial Weight}}{\text{Initial Weight}} \quad 100 = \text{Wt. Increase} \qquad (3)$$

4.19 Reparability. Two panels, 0.040 by 2.75 by 6 inches, of aluminum alloy conforming to QQ-A-250/12 treated per MIL-C-5541 shall be primed and cleaned per 4.6.1. A continuous layer of sealant 0.125 ± 0.015 inch thick shall be applied on 5 inches of the panels. After standard cure, the panels shall be immersed for 7 days at 120° ± 2°F in a 1-quart jar containing 3 percent NaCl solution. The panels shall be dried 24 to 48 hours at standard conditions, cleaned per 4.6.1, and made up into peel specimens as described in 4.14. After standard cure, the panels shall be tested (without further immersion) for peel strength as described in 4.14.

4.20 Crazing effects. A plastic strip, MIL-P-5425B dimensions as shown in figure 4, shall be solvent cleaned with naphtha, TT-N-95 Type II, and set up as a cantilever beam underload in the test apparatus, figure 4, with hole end outboard. After the strip has been loaded to 2000 psi fiber stress, the upper surface of the strip, directly over the fulcrum and extending for 1 inch on both sides, shall be coated with approximately 1/8

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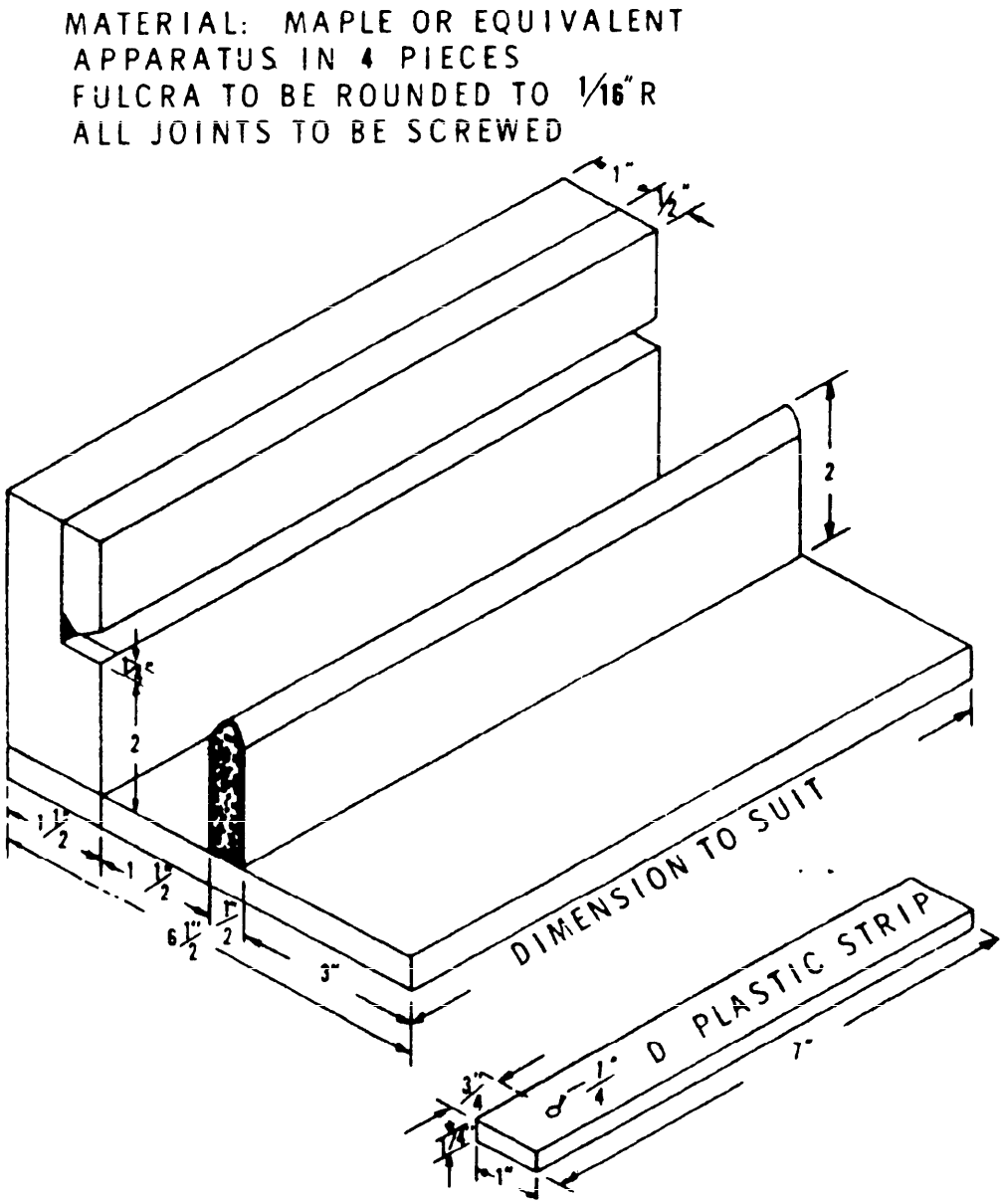


Figure 4. Crazing test jig

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inch thickness of freshly activated sealant. The strip shall remain under stress for 24 hours. After this period, the weight shall be removed and the strip carefully cleaned with an acrylic spatula. Examination shall disclose no evidence of crazing, cracking, or other attack. The load necessary to produce the fiber stress required may be determined by using the following formula:

$$P = \frac{Sbd^2}{6L}$$ (4)

Wherein S = 2,000 psi stress

P = Load (to nearest 0.05 lbs.)

b = Width of plastic strip, within 0.001 inch

L = Distance between fulcrum and the point where the load is applied (4 inches) within 0.05 inch.

d = Thickness of plastic strip (measured by micrometer to ± 0.001 inch).

4.21 Corrosion. One aluminum panel conforming to QQ-A-250/12 and measuring 0.040 by 2.75 by 6 inches in size shall be primed with the manufacturer's recommended primer and two parallel sealing compound fillets shall be applied approximately 1 inch apart. The fillets shall be applied 3/16 inch thick by 3/4 inch wide by 5 inches long and shall extend to within 1/2 inch of the edges of the panel. The panel shall have a 14-day cure. At least 48 hours before the cure time has expired, the panel shall be dipped in a sealing compound topcoat conforming to MIL-S-4383 and the cure allowed to continue. At the end of the 14-day curing period, the fillets shall be cut down to 1/16 inch and immersed vertically for 20 days at 140°F in a covered glass vessel containing a 3.0 percent aqueous sodium chloride solution. Two inches of the panel shall be immersed and the remainder shall be exposed to the vapors. The panel shall be inspected for conformance with requirements of 3.5.14.

4.22 Resistance to paint strippers. Two 1/8 inch thick by 1 inch wide by 4 inch long specimens that are taken from slabs used for Tensile and Elongation data shall be used for this test. After the standard cure one piece shall be immersed in Turco 5292 Paint Stripper and the other piece shall be immersed in EZ 19A stripper for 1 hour at room temperature (77° ± 2°F). At the end of this time, the specimens shall be removed, washed for about 5 minutes in lukewarm water and then placed in a 120°F oven for 24 hours. After removal from oven and cooling to room temperature, the hardness shall be measured in accordance with ASTM D2240.

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4.23 Accelerated storage stability. A full, tightly closed, 1-quart container of the base compound, and a full, tightly closed container of the curing agent, shall be stored for 14 days at 120° ± 2°F. After cooling 77° ± 2°F for at least 24 hours, tests shall be conducted in accordance with 4.10, 4.11, and 4.13. Four peel panels in accordance with 4.14.1 sprayed with manufacturer's recommended primer, shall be prepared and tested in accordance with 4.14.

4.24 Free diisocyanate. Immediately upon completion of mixing, the free diisocyanate shall be determined in accordance with USAF Specification Bulletin No. 535.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.3).

5.1.1 Level A

5.1.1.1 Unit packaging. The base compound and the curing agent shall each be packaged in its own container. Each curing agent container shall be packaged with one base compound container in a manner that will prevent accidental separation but will permit easy separation for mixing purposes. The ratio of the quantity contained in base container to the quantity contained in the curing agent container shall be the same as the recommended mixing ratio of the base compound and curing agent.

5.1.1.1.1 Individual containers. The base compound shall be furnished in 1/2-pint, 1-pint, 1-quart, or 1-gallon metal cans, as specified in the contract or order. Metal cans shall conform to type V, class 2 of PPP-C-96, except that tinplate cans with paper labels may be used, unless specifically prohibited. The base compound contained in each size container shall be as follows:

Size of container	Amount of base compound	Volume tolerance
1/2 pint	6 fluid ounces	± 1/8 fluid ounce
1 pint	12 fluid ounces	± 1/4 fluid ounce
1 quart	24 fluid ounces	± 1/2 fluid ounce
1 gallon	96 fluid ounces	± 2 fluid ounces

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The air in the top of the base compound containers shall be replaced with nitrogen gas immediately prior to closing the containers. The curing agent shall be furnished in glass jars or in suitable plastic containers approved by the procuring activity. The jars or plastic containers, as applicable, shall have vertical, smooth inside walls, and no internal projections nor internal lips exceeding 1/16 inch. The glass jars shall be closed with enameled metal or plastic continuous thread screw caps having a nonabsorbent lining material. Caps shall be tightened adequately and further sealed with cellulose bands, or equivalent.

5.1.1.1.2 Sectional-type containers. The base compound and curing agent shall be furnished in sectional-type 2-1/2-ounce or 6-ounce nonmetal containers, conforming to MIL-P-38714, as specified in the contract or order. The total content of base compound and curing agent contained in each sectional-type container shall be as follows:

Size of container	Total content (base and curing)	Volume tolerance
2-1/2 ounce	2 fluid ounces	± 1/8 fluid ounce
6 ounce	3-1/2 fluid ounces	± 1/8 fluid ounce

5.1.2 Level C. Packaging shall be in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.3).

5.2.1 Level A. The sealing compound shall be packed in overseas-type shipping containers in accordance with the requirements of the appendix to PPP-C-96.

5.2.2 Level B. The sealing compound shall be packed in domestic-type shipping containers in accordance with the requirements of the appendix to PPP-C-96.

5.2.3 Level C. The sealing compound shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking of shipment. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. Marking shall include specification number, applicable class and dash number, name of manufacturer, date of manufacture (month and year), batch number, mixing instructions, and recommended storage temperature.

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6. NOTES

6.1 Intended use. The sealing compound covered by this specification is intended for sealing pressure and weather cavity areas in a temperature range of -67°F to +200°F.

6.2 Warning. These materials are not intended for faying surface sealing or for wet installing fasteners since the fillers used to obtain low density may reduce structural fatigue strength when used to seal between metal surfaces.

6.3 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Class of sealing compound and dash number (see 1.2 and 1.2.1).

c. Quantity desired.

d. Size of container (see 5.1.1.1.1 and 5.1.1.1.2).

e. Applicable levels of packaging and packing (see 5.1 and 5.2).

f. Special marking if required (see 5.3).

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to have the products that they propose to offer the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Air Force Materials Laboratory, ATTN: LAE, Wright-Patterson AFB, Ohio 45433, and information pertaining to qualification of products may be obtained from that activity.

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Preparing activity:
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